Protocol details: minimal changes

Document 5 | Four small, appendix-grounded edits

Exact source text is quoted verbatim (LaTeX where applicable). Page references identify the saved before.pdf snapshot, not a newly compiled manuscript. All edits below are NOT APPLIED.

The top-50 candidate pool and five-versus-ten downstream budgets already appear in the main text. The proposals clarify their interpretation; they do not claim these numbers are missing.

1. Correct the split description — NOT APPLIED

Source: sections/experiments.tex:12–12 | before.pdf p. 4

OLD — exact source fragment:

We split each dataset into train, validation, and test splits with ratios of 70\%, 15\%, 15\%, respectively. To evaluate the learning capabilities of baselines, we further filter and balance splits such that each split has at least one task that share the same required skill.

NEW — exact replacement:

After iterative rare-skill filtering, 4,987 of 5,400 tasks remain. A constrained global 70\%/15\%/15\% split yields 3,491/748/748 task-disjoint train/validation/test instances, with all 306 retained gold skills represented in each partition; this tests held-out-task rather than unseen-skill generalization (Appendix~\ref{app:data-splits}).

Readable proposal (same wording, references expanded):

After iterative rare-skill filtering, 4,987 of 5,400 tasks remain. A constrained global 70%/15%/15% split yields 3,491/748/748 task-disjoint train/validation/test instances, with all 306 retained gold skills represented in each partition; this tests held-out-task rather than unseen-skill generalization (Appendix C.1).

Why: Replace only the last two sentences of the paragraph. The current per-dataset ratio is inaccurate; the appendix explicitly gives a global constrained allocation and shared skill vocabulary.

Grounding: sections/appendix/data_splits.tex:3 (snapshot p. 16); sections/appendix/data_splits.tex:5 (snapshot p. 16); sections/appendix/data_splits.tex:7 (snapshot p. 16)

Candidate pool and downstream budget

2. Separate candidates from supplied skills — NOT APPLIED

Source: sections/experiments.tex:47–47 | before.pdf p. 5

OLD — exact source fragment:

The skill routing baselines are conditioned on the top 50 retrieved skills and use some optimization method to improve performance.

NEW — exact replacement:

The skill routing baselines use a retrieved candidate pool of 50 skills, distinct from the smaller downstream skill budget.

Why: The top-50 pool is already in the current main text; this is a distinction/clarity edit, not promotion of an absent fact. Leave the immediately following SkillDAG exception unchanged.

Grounding: sections/appendix/implementation_settings.tex:3 (snapshot p. 17)

3. Qualify the downstream comparison — NOT APPLIED

Source: sections/experiments.tex:47–47 | before.pdf p. 5

Existing anchor — retain unchanged (exact source):

In downstream tasks, we use k = 5 skills for SkillRouter and $\mathrm{RouterBSI}_{\mathrm{top}k}$, and we set k = 10 for SkillDAG because k = 5 would match the retriever's performance.

INSERT immediately after anchor:

The strict RouterBSI variant supplies the variable-size predicate-selected set, so the downstream comparisons are not matched for supplied skill count.

Why: Keep the existing five-versus-ten budgets and SkillDAG rationale unchanged. Add only the strict variant and the unequal-budget qualification.

Grounding: sections/appendix/implementation_settings.tex:3 (snapshot p. 17); sections/appendix/error_downstream_results.tex:2 (snapshot p. 28); sections/appendix/procedure_routing.tex:18 (snapshot p. 11)

Uncertainty reporting

4. Name the statistic in Figure 5 — NOT APPLIED

Source: figs/mbr_replay_test_ndcg@1_std/description.text:1–1 | before.pdf p. 7

OLD — exact source fragment:

The lines and areas represent the standard mean and variation, respectively across three runs starting from the same initial predicates at iteration 0.

NEW — exact replacement:

Across three runs starting from the same initial predicates at iteration 0, lines show mean test nDCG@1 and shaded bands show plus/minus one population standard deviation, not confidence intervals.

Why: Small caption-only replacement gives the statistic and aggregation unit while preserving the author-confirmed shared initialization.

Grounding: sections/appendix/trajectory_table.tex:22 (snapshot p. 24); sections/appendix/ablation_components.tex:21 (snapshot p. 23)

Shared initialization is preserved exactly as clarified by the author. The change identifies population standard deviation across three runs; it neither introduces confidence intervals nor asserts statistical significance.